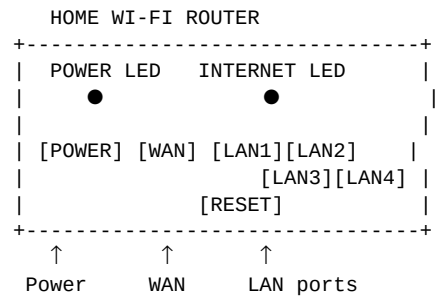


Day 2 – Routers and Switches

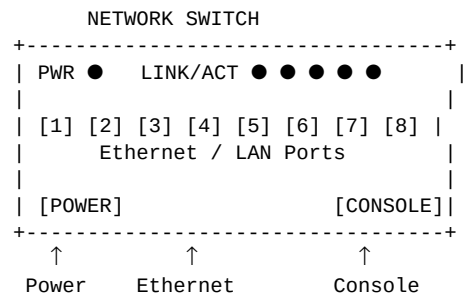
Basic Functions, Ports, LEDs and Practical Network Setup

Task 1: Labelled Diagrams

Typical Home Wi-Fi Router



Typical Network Switch



Router LEDs show power, Internet/WAN and LAN activity. Switch LEDs normally show power and link/activity for each port.

Task 2: Three Key Differences

Point	Router	Switch
Main function	Connects different networks and provides Internet access.	Connects devices within the same LAN.
Address used	Uses IP addresses to route packets.	Mainly uses MAC addresses to forward frames.
Traffic / network	Moves traffic between different networks, such as LAN and Internet.	Forwards traffic between devices on the same local network.

Task 3: Ports and LED Identification

Router

- **Power port:** Usually at the back; connects the power adapter.
- **Ethernet ports:** WAN/Internet port connects to the modem/ISP; LAN ports connect wired devices.
- **LED indicators:** Usually on the front; show power, Internet and LAN/Wi-Fi activity.

Switch

- **Power port:** Usually at the back or side; connects the power supply.
- **Ethernet ports:** Multiple LAN ports connect PCs, printers and other wired devices.
- **LED indicators:** Usually near the ports; show power and link/activity for connected ports.

Task 4: Small Gaming Café Scenario

For a small gaming café, I would connect the **router first to the ISP line**. The router receives the Internet connection and manages the network. Then I would connect one LAN port of the router to the network switch. The switch would then connect to all **8 gaming PCs** using Ethernet cables.

```
ISP
|
[ROUTER]
|
```

```
[SWITCH]
|-- PC 1
|-- PC 2
|-- PC 3
|-- PC 4
|-- PC 5
|-- PC 6
|-- PC 7
|-- PC 8
```

This setup is suitable for IPL streaming and multiplayer gaming because the router handles Internet access and network routing, while the switch provides multiple wired connections for the gaming PCs.

Conclusion: A router connects different networks and uses IP addresses, while a switch connects devices in the same LAN and mainly uses MAC addresses. In a gaming café, the router should connect to the ISP first, followed by the switch and gaming PCs.